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Kinematics of the interstellar medium using *Gaia*: A catalogue of 102 young stellar object–molecular cloud associations within 3.5 kpc of the Sun with 3D velocities

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ABSTRACT

Kinematic information is crucial for understanding the evolution of complex systems, such as interstellar gas. Obtaining full 3D kinematic information is a vital final step for modelling and interpretation. Molecular clouds (MCs) are nurseries where stars are born. Stars at a very early stage, such as young stellar objects (YSOs), inherit the spatial and kinematic structure of the gas patches in which they originate. In this paper, we combine measurements of radial velocities towards the gas and the kinematic information of YSOs from *Gaia* DR3 to derive 3D velocities of a sample of YSO–MC complexes at $d \lesssim 3.5$ kpc from the Sun. We find that the molecular interstellar medium traced by the YSO–MC complexes generally follows Galactic rotation, with an additional peculiar velocity of 8.6 km s^{-1}. The random motion of these complexes in the Galactic x-y-plane is more energetic than motion along the z-direction. A catalogue containing the 3D velocities of the YSO–MC complexes in various reference frames is available, and the distances and 3D velocities of well-known MCs are presented. Our results establish the foundation for exploring the interplay between the Galaxy, the molecular interstellar medium, and star formation.

Key words: Galaxies: ISM – ISM: structure – ISM: clouds – Stars: formation – ISM: kinematics and dynamics.

1 INTRODUCTION

Molecular clouds (MCs) are the birthplace of stars. They contain the most cold, molecular gas in our Galaxy (Dame, Hartmann & Thaddeus 2001). These clouds are not isolated objects in galaxies but rather participate in the material cycle in galaxies. The formation of MCs is a direct consequence of the evolution of the Galactic disc (e.g. Pringle, Allen & Lubow 2001). During the cloud lifetime, Galactic-scale processes constantly affect cloud evolution by setting the boundary conditions where mass and kinematic energy are injected. The motion of MCs is affected by many physical processes such as Galactic shear (Li et al. 2013; Goodman et al. 2014; Ragan et al. 2014; Dobbs 2015; Wang et al. 2015; Li & Zhang 2020; Li, Zhou & Chen 2022), stellar feedback (Ceverino & Klypin 2009), and spiral density waves (Lin, Yuan & Shu 1969; Dobbs & Baba 2014). In addition to the differential rotation caused by gravity from dark matter, stars and MCs still contain a significant amount of random motion.

Kinematic information is crucial for understanding the interplay between the clouds and the Galaxy, and full 3D kinematic measurements of MCs are vital for understanding their evolution in the Galaxy. Earlier studies have identified many MCs from CO emission maps, from which the properties of clouds such as size, mass, radial velocity, velocity dispersions, and CO luminosities can be measured (Scoville & Solomon 1975; Solomon et al. 1987; Roman-Duval et al. 2010; Chen et al. 2020). The kinematic measurements derived using velocity-resolved CO measurements or other molecular lines are limited to 1D velocity along the radial direction.

Young stellar objects (YSOs) are stars in the earliest stages of evolution and are often used as tracers of MCs. Based on their near- to mid-infrared spectral index, YSOs are classified into Class 0, I, II, and III (Lada 1987). Through this sequence, young stars are classified from being deeply embedded in protostellar cores in the Class 0 stage to being surrounded by an envelope (Class I) or circumstellar disc (Class II), and then to have the spectral energy distribution of stellar photospheres in the Class III stage (Dunham et al. 2014). Being embedded in MCs or bearing discs or envelopes, YSOs exhibit infrared excesses, which are widely used to identify them. In addition, other observational features, for example outflows and jets driven by the accretion process (Andre, Ward-Thompson & Barsony 1993; Arce et al. 2007; Frank et al. 2014), and X-ray emission generated from the interaction between the disc and the magnetic field (Neuhäuser 1997; Feigelson & Montmerle 1999), are also used to characterize YSOs.

With the increasing number of YSOs identified since the launch of *Spitzer*, *Herschel*, and the *W ide-field Infrared Survey Explorer* (*WISE*) (Evans et al. 2003, 2009; André et al. 2010; Fischer et al. 2016), studies of their spatial distribution and kinematics have gained significant momentum (Dunham et al. 2014). Before MCs are dispersed or expelled, YSOs remain spatially associated with them, having similar locations. Precise parallax measurements of YSOs have enabled accurate determinations of distances and positions for many MCs (Dzib et al. 2018; Marton et al. 2023; Zhang 2023). YSOs are also thought to inherit the velocities of their parent MCs. Earlier studies comparing CO gas and YSO radial velocities in

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nearby clouds confirmed their dynamical association (Covey et al. 2006). Using high-resolution optical spectra from young stars in the Orion B molecular cloud, Kounkel et al. (2017) found that those in NGC 2068 and L1622 exhibit radial velocities similar to those of the molecular gas. Similarly, Großschedl et al. (2021) reported a close agreement between the mean radial velocities of CO gas and YSOs in the Orion molecular cloud. In more massive environments, such as the OB clusters in Cygnus, Quintana & Wright (2022) demonstrated that the 3D velocity dispersion of expanding clusters follows Larson's law, indicating a connection to the turbulent dynamics of their parent clouds. Yang et al. (2025) found that radial velocity differences between CO gas and YSOs are typically less than 1.4 km s^{-1} in five nearby star-forming regions (Orion A, Orion B, Perseus, Taurus, and λ Orionis). Consequently, YSO kinematics are now widely utilized to investigate the motions and evolution of MCs, including their expansion, structural evolution, and turbulence (Dzib et al. 2018; Kounkel et al. 2018; Ha et al. 2021; Zhou, Li & Chen 2022; Swiggum, Alves & D'Onghia 2025). In this work, we utilize YSO associations linked to MCs, defined as YSO–MC complexes. These associations enable us to study the 3D velocity structure of MCs, supplemented by 2D stellar velocity data from *Gaia* DR3.

The *Gaia* satellite (Gaia Collaboration et al. 2016) provides high-precision astrometric measurements, enabling the determination of stellar distances, velocities, spectral types, and stellar parameters. The latest data release, *G aia* DR3, includes observations of over 1.8 billion sources, offering improved parallaxes and proper motions, particularly for faint stars, and a larger number of radial velocity measurements (Gaia Collaboration et al. 2023). Full astrometric solutions are available for approximately 1.46 billion sources. Notably, the measurements of YSOs are especially valuable, as they can be used to trace the motion of the gas associated with them. In our previous work, we studied the kinematic properties of a sample of YSO associations and their gas counterparts (Zhou et al. 2022, 2024). Based on their gas contents, we select a sample of YSO–MC complexes in which the gas and a group of YSOs are associated with each other. For these complexes, the full 3D velocities can be obtained by combining the radial velocities measured towards the gas with astrometry measurements towards the YSOs provided by the *Gaia* satellite. The scientific potential of this approach has been demonstrated in our previous papers (Li et al. 2022). In this paper, we present a catalogue of YSO–MC complexes with full, 3D velocities in various reference frames located at $d \lesssim 3.5$ kpc from the Sun. We present our calculations with full details and discuss the peculiar velocities of the YSO–MC complexes.

2 DERIVATION OF 3D VELOCITIES

2.1 Assumptions on the Galactic dynamics

Given the close relationship between the kinematics of YSOs and MCs found in the references in Section 1, we made the assumption that the YSOs keep the velocity inherited from their parent MCs. We limited ourselves to certain kinds of YSO associations, where YSOs have a similar spatial distribution to the gas, indicating that they are young enough and associated with the gas. We took the mean proper motion in the l and b directions of the YSO associations as the velocity of the clouds. Then, the 2D velocities can be combined with the radial velocity of the gas to fully describe the cloud motion.

2.2 YSO associations with *Gaia* kinematic measurements

A sample containing 102 YSO associations is taken from our last work (Zhou et al. 2022), where the YSOs used are Class I and Class II type (Marton et al. 2016). These YSO associations are extracted from their density map in l -b-log(d) space using the `Dendrogram` method (Rosolowsky et al. 2008).[1] The 102 associations contain large structures and denser substructures.

The YSO associations are classified into different types based on their relation with the molecular gas in their vicinity (Zhou et al. 2024). The 102 associations used in this work are all gas-rich, including the Type 1 direct association (where most of the YSOs are closely associated with a cloud), Type 2 close association (where some of the YSOs are associated with the clouds), and the Type 3 bubble association (YSOs are located within bubble gas structures). As these are objects associated with the molecular gas, the radial velocity can be obtained reliably using gas tracers. In this paper, we call these objects YSO–MC complexes. To update the location and velocity information of the member YSOs, we cross-matched them with the *Gaia* DR3 catalogue, which provides more accurate photometric and kinematic measurements (Gaia Collaboration et al. 2023). For each YSO–MC complex, the proper motion (pmra and pmdec) and 3D location (l, b, d) are obtained from *Gaia* DR3, and the proper motions in right ascension (RA) and declination (Dec.) are later transformed into pm$l$ and pm$b$ in Galactic coordinates using GALPY (Bovy 2015).

2.3 Radial velocity measurements from CO

The radial velocity towards each cloud is derived from ^{12}CO emission data (Dame et al. 2001). This composite CO data set towards the Milky Way was assembled from seven individual surveys. It has an angular resolution ranging from 9′ to 18′, with a velocity resolution of 1.3 km s^{-1}. The matching between YSO associations and CO clouds has already been performed, and details can be found in Zhou et al. (2024), where the radial velocities as well as their uncertainties were derived from the line profile of the CO emissions. We used several Gaussian components to fit the CO line profile and derived the central velocities and standard deviation as the velocity and its uncertainty. By checking the spatial distribution of the gas and YSOs visually, we chose the best-matching CO component associated with the YSO association. The central velocity and the uncertainty of the best-matching component are taken as the radial velocity and its error.

3 DERIVING 3D VELOCITIES

The aim of this paper is to derive 3D velocities towards our sample YSO–MC complexes by combining the CO radial velocity with the transverse velocities obtained by the *Gaia* satellite. For historical reasons, past measurements were performed in different frames whose definitions can sometimes be vague. Thus, the first step of our velocity conversion is to convert all measurements to the barycentric frame. The measured velocities are projected into different frames for convenience. These details will be discussed in this section. A list of all the reference frames can be found in Appendix A, and the conversion process is illustrated in Fig. 1.

[1] http://www.dendrograms.org/.

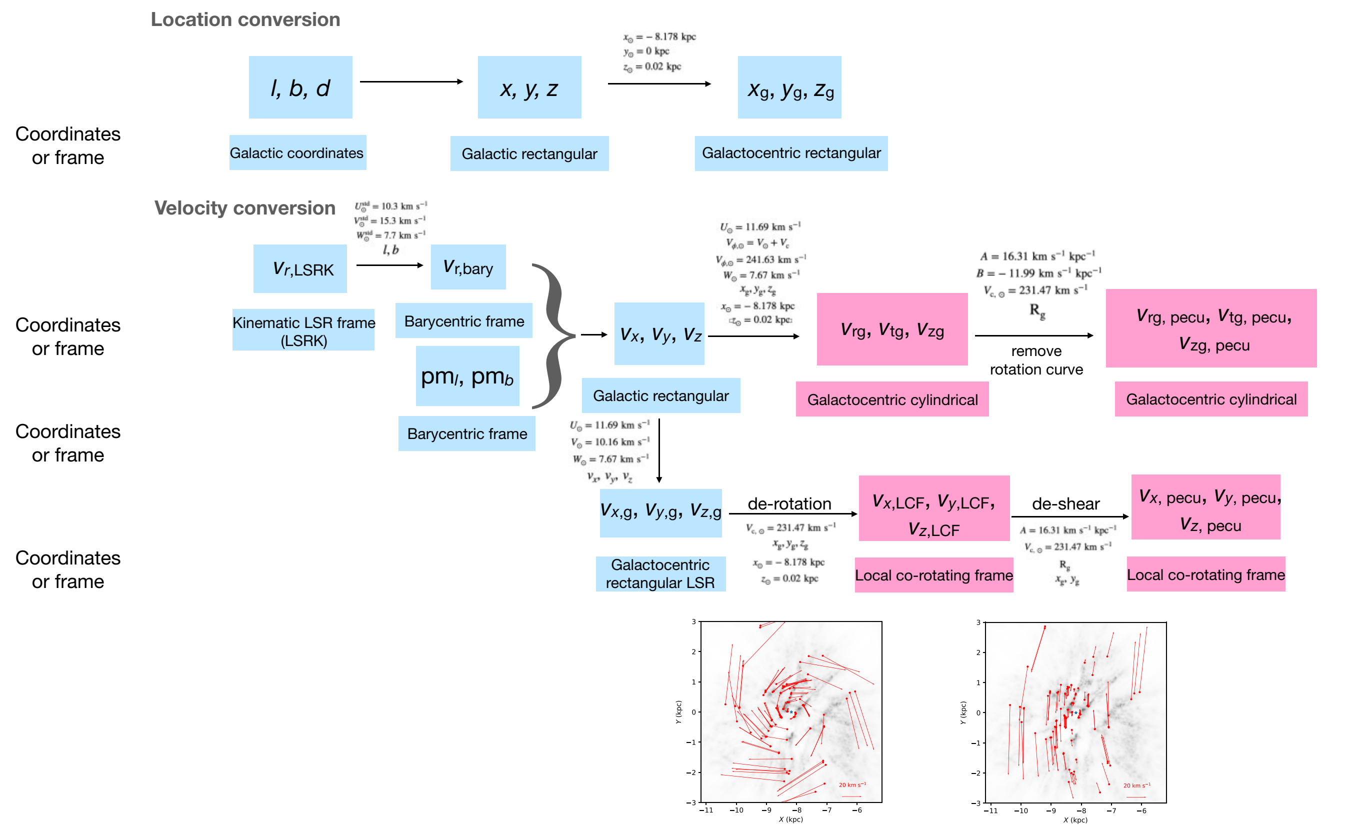

Figure 1. A diagram of the location and velocity conversion. We present velocity and location conversions, with the coordinates or reference frames written below. All the solar location, velocity, and association parameters are noted and given. The subplot below 'de-rotation' shows the rotation field, and the one below 'de-shear' shows the shear field.

3.1 Location conversions

We adopt two sets of locations for the YSO–MC complexes, as follows.

(i) (x, y, z) in Galactic rectangular coordinates. In these coordinates, the x-axis points from the Sun to the Galactic Centre, and the y-axis is perpendicular to the x-axis and points in the direction that follows the Galactic rotation. The z-axis points to the North Galactic Pole. The Sun is located at (0, 0, 0). The Galactic Centre is located at (8.178 kpc, 0, 0) (GRAVITY Collaboration et al. 2019). (x, y, z) is related to (l, b, d) by $x = d\cos(b)\cos(l)$, $y = d\cos(b)\sin(l)$, $z = d\sin(b)$, where l and b are the Galactic longitude and latitude, and d is the distance from the Sun.

(ii) (x_g, y_g, z_g) in Galactocentric rectangular coordinates. In these coordinates, the Sun is located at (–8.178 kpc, 0 kpc, 0.02 kpc) (Bennett & Bovy 2019; GRAVITY Collaboration et al. 2019), and the Galactic Centre is located at (0, 0, 0). The x_g-axis points from the Sun to the Galactic Centre, the y_g-axis points from the Galactic Centre to $l = 90^\circ$, and the z_g-axis points roughly to the North Galactic Pole. To convert from (x, y, z) to (x_g, y_g, z_g), one needs the location of the Sun, $x_\odot$, and the solar height above the Galactic midplane, $z_\odot$. We adopt $x_\odot = -8.178$ kpc and $z_\odot = 0.02$ kpc (Bennett & Bovy 2019; GRAVITY Collaboration et al. 2019).

These two sets of coordinates are shown in Fig. 2. The angle between the Galactocentric rectangular coordinates and the Galactic rectangular coordinates is $\arctan(z_\odot/x_\odot)$.

3.2 Velocity conversions

The computation of 3D velocities consists of a few key steps, as follows.

(i) Convert radial velocity measurements from the kinematic local standard of rest (LSR) frame to the barycentric frame, where the *Gaia* astrometry measurements are represented.
(ii) Derive 3D velocities in the barycentric frame.
(iii) Convert velocity measurements to other frames.

3.2.1 *Converting CO velocities into the barycentric frame*

The proper motions and locations of YSO–MC complexes are provided by *Gaia* DR3 and are measured in the barycentric frame, whose velocity is measured with respect to the centre of gravity of the Solar System.

The radial velocity of each CO cloud is derived from the best-fitting CO emission and has already been converted to kinematic local standard of rest (LSRK) coordinates.[2] To derive the 3D velocity, we first convert this measurement back to the barycentric frame. This conversion requires a modification of the standard solar motion in the line of sight. We use the following equation from Reid et al. (2009):

$$v_{\mathrm{r,bary}} = v_{\mathrm{r,LSRK}} - (U_\odot^{\mathrm{Std}}\cos l + V_\odot^{\mathrm{Std}}\sin l)\cos b - W_\odot^{\mathrm{Std}}\sin b, \quad (1)$$

where $v_{\mathrm{r,bary}}$ is the radial velocity in the barycentric frame, l and b are the mean Galactic longitude and latitude of the YSO–MC complex,

[2] https://docs.astropy.org/en/stable/api/astropy.coordinates.LSRK.html

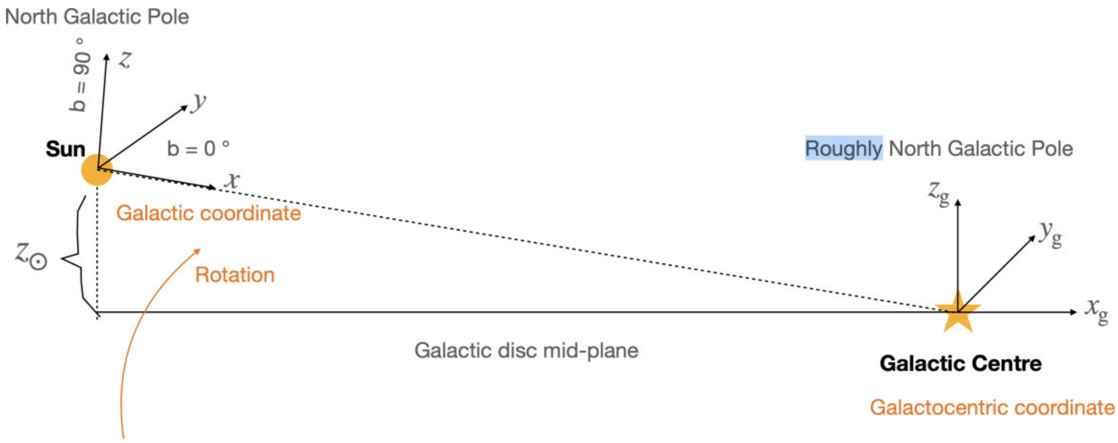

Figure 2. A comparison between Galactic rectangular and Galactocentric rectangular coordinates. The orange star and the orange circle mark the location of the Galactic Centre and the Sun, respectively. The orange curve shows the direction of Galactic rotation.

Table 1. 3D velocities for YSO–MC complexes. The locations, proper motions, radial velocities, and 3D velocities and errors in various coordinates are presented in the table. Only a small portion of the table is published here. The full version is available online and also at https://doi.org/10.5281/zenodo.16364877.

YSO–MC ID	0	1	2	3	4	5	6	7	8	9	...
l (°)	171.67	169.56	168.68	174.47	−21.80	−23.58	−20.33	−7.12	−6.74	−6.80	...
b(°)	−15.46	−15.70	−15.88	−15.00	8.99	8.56	9.35	19.48	18.91	17.01	...
d (kpc)	0.13	0.13	0.13	0.14	0.16	0.16	0.16	0.14	0.14	0.14	...
pm$_l$ (mas yr^{-1})	21.85	23.19	23.98	19.80	−23.40	−23.58	−23.23	−24.13	−24.14	−24.10	...
pm$_b$ (mas yr^{-1})	−9.69	−10.83	−11.19	−7.35	−10.55	−10.55	−10.58	−10.12	−10.72	−11.54	...
$v_{r,\ \mathrm{bary}}$ (km s^{-1})	14.91	15.64	15.41	15.65	0.16	−0.53	−0.34	−7.82	−6.57	−5.10	...
$v_{r,\ \mathrm{err}}$ (km s^{-1})	1.30	1.30	1.30	1.30	1.95	1.30	2.60	1.30	0.65	0.00	...
x_g (kpc)	−8.30	−8.30	−8.30	−8.32	−8.03	−8.04	−8.03	−8.05	−8.05	−8.05	...
y_g (kpc)	0.02	0.02	0.02	0.01	−0.06	−0.06	−0.05	−0.02	−0.02	−0.02	...
z_g (kpc)	−0.01	−0.02	−0.02	−0.02	0.04	0.04	0.05	0.07	0.07	0.06	...
v_x (km s^{-1})	−14.60	−15.64	−15.58	−15.05	−5.22	−6.43	−5.18	−7.07	−5.77	−4.51	...
v_y (km s^{-1})	−11.73	−11.90	−11.94	−12.03	−16.79	−16.33	−16.74	−15.44	−15.44	−15.46	...
v_z (km s^{-1})	−9.84	−10.77	−10.85	−8.86	−7.78	−7.84	−7.92	−9.01	−8.86	−8.77	...
$v_{x,\ \mathrm{err}}$ (km s^{-1})	1.24	1.23	1.23	1.25	1.79	1.18	2.41	1.22	0.61	0.02	...
$v_{y,\ \mathrm{err}}$ (km s^{-1})	0.25	0.31	0.29	0.26	0.72	0.53	0.90	0.24	0.20	0.21	...
$v_{z,\ \mathrm{err}}$ (km s^{-1})	0.36	0.37	0.36	0.35	0.31	0.21	0.43	0.44	0.23	0.12	
v_{rg} (km s^{-1})	3.45	4.61	4.59	3.75	−8.08	−6.98	−8.02	−5.06	−6.34	−7.60	...
v_{tg} (km s^{-1})	−229.90	−229.72	−229.67	−229.60	−224.79	−225.25	−224.84	−226.18	−226.17	−226.15	...
v_{zg} (km s^{-1})	−2.13	−3.06	−3.14	−1.16	−0.10	−0.15	−0.23	−1.33	−1.17	−1.09	...
$v_{\mathrm{rg,\ err}}$ (km s^{-1})	1.24	1.23	1.23	1.25	1.79	1.18	2.41	1.22	0.61	0.02	...
$v_{\mathrm{tg,\ err}}$ (km s^{-1})	0.25	0.31	0.29	0.26	0.72	0.53	0.90	0.24	0.20	0.21	...
$v_{\mathrm{zg,\ err}}$ (km s^{-1})	0.36	0.37	0.36	0.35	0.31	0.21	0.43	0.44	0.23	0.12	...
$v_{\mathrm{rg,\ pecu}}$ (km s^{-1})	3.45	4.61	4.59	3.75	−8.08	−6.98	−8.02	−5.06	−6.34	−7.60	...
$v_{\mathrm{tg,\ pecu}}$ (km s^{-1})	−1.03	−1.21	−1.27	−1.28	−7.31	−6.83	−7.27	−5.86	−5.86	−5.89	...
$v_{\mathrm{zg,\ pecu}}$ (km s^{-1})	−2.13	−3.06	−3.14	−1.16	−0.10	−0.15	−0.23	−1.33	−1.17	−1.09	...
$v_{x,\ \mathrm{g}}$ (km s^{-1})	−2.94	−3.97	−3.92	−3.38	6.45	5.24	6.49	4.60	5.90	7.16	...
$v_{y,\ \mathrm{g}}$ (km s^{-1})	−1.57	−1.74	−1.78	−1.87	−6.63	−6.17	−6.58	−5.28	−5.28	−5.30	...
$v_{z,\ \mathrm{g}}$ (km s^{-1})	−2.13	−3.06	−3.14	−1.16	−0.10	−0.15	−0.23	−1.33	−1.17	−1.09	...
$v_{x,\ \mathrm{g,\ err}}$ (km s^{-1})	1.24	1.23	1.23	1.25	1.79	1.18	2.41	1.22	0.61	0.02	...
$v_{y,\ \mathrm{g,\ err}}$ (km s^{-1})	0.25	0.31	0.29	0.26	0.72	0.53	0.90	0.23	0.20	0.21	...
$v_{z,\ \mathrm{g,\ err}}$ (km s^{-1})	0.36	0.37	0.36	0.35	0.31	0.21	0.43	0.44	0.23	0.12	...
$v_{x,\ \mathrm{LCF}}$ (km s^{-1})	−3.46	−4.62	−4.61	−3.76	8.09	7.00	8.03	5.06	6.34	7.60	...
$v_{y,\ \mathrm{LCF}}$ (km s^{-1})	−5.14	−5.29	−5.26	−5.76	−2.53	−2.15	−2.42	−1.54	−1.56	−1.56	...
$v_{z,\ \mathrm{LCF}}$ (km s^{-1})	−2.13	−3.06	−3.14	−1.16	−0.10	−0.15	−0.23	−1.33	−1.17	−1.09	...
$v_{x,\ \mathrm{pecu}}$ (km s^{-1})	−3.45	−4.61	−4.60	−3.75	8.13	7.04	8.07	5.07	6.35	7.61	...
$v_{y,\ \mathrm{pecu}}$ (km s^{-1})	−1.02	−1.20	−1.25	−1.27	−7.25	−6.78	−7.21	−5.85	−5.85	−5.87	...
σ_{2D} (km s^{-1})	1.94	1.88	1.34	2.05	1.92	2.72	0.82	1.85	1.71	1.67	...

Table 2. 3D velocities of some well-known MCs. The locations, proper motions, radial velocities, and 3D velocities and errors of Taurus, Lupus, Ophiuchus, Chamaeleon, Perseus, Orion, and λ Orion at various coordinates are presented.

Cloud name	Taurus	Lupus	Ophiuchus	Chamaeleon	Perseus	Orion	λ Orion
l (°)	171.67	−21.80	−7.12	−56.42	158.45	−151.64	−165.46
b(°)	−15.46	8.99	19.48	−14.67	−20.74	−18.44	−11.39
d (kpc)	0.13	0.16	0.14	0.20	0.29	0.39	0.40
pm_l (mas yr^{-1})	21.85	−23.40	−24.13	−20.54	11.31	0.87	2.30
pm_b (mas yr^{-1})	−9.69	−10.55	−10.12	−6.61	−2.71	0.49	0.27
$v_{r,\ \mathrm{bary}}$ (km s^{-1})	14.91	0.16	−7.82	10.06	10.30	23.14	25.48
$v_{r,\ \mathrm{err}}$ (km s^{-1})	1.30	1.95	1.30	1.30	1.30	2.60	1.95
x_g (kpc)	−8.30	−8.03	−8.05	−8.07	−8.43	−8.51	−8.56
y_g (kpc)	0.02	−0.06	−0.02	−0.16	0.10	−0.18	−0.10
z_g (kpc)	−0.01	0.04	0.07	−0.03	−0.08	−0.10	−0.06
v_x (km s^{-1})	−14.60	−5.22	−7.07	−11.43	−13.48	−18.80	−23.18
v_y (km s^{-1})	−11.73	−16.79	−15.44	−17.40	−11.51	−11.99	−10.53
v_z (km s^{-1})	−9.84	−7.78	−9.01	−8.51	−7.16	−6.45	−4.52
$v_{x,\ \mathrm{err}}$ (km s^{-1})	1.24	1.79	1.22	0.71	1.14	2.17	1.85
$v_{y,\ \mathrm{err}}$ (km s^{-1})	0.25	0.72	0.24	1.05	0.56	1.17	0.49
$v_{z,\ \mathrm{err}}$ (km s^{-1})	0.36	0.31	0.44	0.33	0.48	0.83	0.39
v_rg (km s^{-1})	3.45	−8.08	−5.06	−4.64	4.54	2.35	8.83
v_tg (km s^{-1})	−229.90	−224.79	−226.18	−224.18	−230.09	−229.74	−231.22
v_zg (km s^{-1})	−2.13	−0.10	−1.33	−0.81	0.55	1.27	3.21
$v_{\mathrm{rg},\ \mathrm{err}}$ (km s^{-1})	1.24	1.79	1.22	0.71	1.14	2.17	1.85
$v_{\mathrm{tg},\ \mathrm{err}}$ (km s^{-1})	0.25	0.72	0.23	1.05	0.56	1.17	0.49
$v_{\mathrm{zg},\ \mathrm{err}}$ (km s^{-1})	0.36	0.31	0.44	0.33	0.48	0.83	0.39
$v_{\mathrm{rg},\ \mathrm{pecu}}$ (km s^{-1})	3.45	−8.08	−5.06	−4.64	4.54	2.35	8.83
$v_{\mathrm{tg},\ \mathrm{pecu}}$ (km s^{-1})	−1.03	−7.31	−5.86	−7.73	−0.28	−0.30	1.40
$v_{\mathrm{zg},\ \mathrm{pecu}}$ (km s^{-1})	−2.13	−0.10	−1.33	−0.81	0.55	1.27	3.21
$v_{x,\ \mathrm{g}}$ (km s^{-1})	−2.94	6.45	4.60	0.24	−1.81	−7.13	−11.50
$v_{y,\ \mathrm{g}}$ (km s^{-1})	−1.57	−6.63	−5.28	−7.24	−1.35	−1.83	−0.37
$v_{z,\ \mathrm{g}}$ (km s^{-1})	−2.13	−0.10	−1.33	−0.81	0.55	1.27	3.21
$v_{x,\ \mathrm{g},\ \mathrm{err}}$ (km s^{-1})	1.24	1.79	1.22	0.71	1.14	2.17	1.85
$v_{y,\ \mathrm{g},\ \mathrm{err}}$ (km s^{-1})	0.25	0.72	0.24	1.05	0.56	1.17	0.49
$v_{z,\ \mathrm{g},\ \mathrm{err}}$ (km s^{-1})	0.36	0.31	0.44	0.33	0.48	0.83	0.39
$v_{x,\ \mathrm{LCF}}$ (km s^{-1})	−3.46	8.09	5.06	4.72	−4.65	−2.12	−8.70
$v_{y,\ \mathrm{LCF}}$ (km s^{-1})	−5.14	−2.53	−1.54	−4.26	−8.53	−11.10	−11.15
$v_{z,\ \mathrm{LCF}}$ (km s^{-1})	−2.13	−0.10	−1.33	−0.81	0.55	1.27	3.21
$v_{x,\ \mathrm{pecu}}$ (km s^{-1})	−3.45	8.13	5.07	4.79	−4.55	−2.35	−8.84
$v_{y,\ \mathrm{pecu}}$ (km s^{-1})	−1.02	−7.25	−5.85	−7.64	−0.23	−0.35	1.29
σ_2D (km s^{-1})	1.94	1.92	1.85	0.82	2.43	3.06	3.19

and ($U_\odot^\mathrm{Std}$, $V_\odot^\mathrm{Std}$, $W_\odot^\mathrm{Std}$) represents the standard solar motion towards the Galactic Centre, $l = 90°$, and the North Galactic Pole. Following Reid et al. (2009), we adopt the standard solar motion values of $U_\odot^\mathrm{Std} = 10.3\,\mathrm{km\,s^{-1}}$, $V_\odot^\mathrm{Std} = 15.3\,\mathrm{km\,s^{-1}}$, $W_\odot^\mathrm{Std} = 7.7\,\mathrm{km\,s^{-1}}$, which are consistent with the values used when reducing radio observations (CASA Team et al. 2022).

3.2.2 *Velocities in the Galactic rectangular coordinate frame*

The next step is to convert velocities to the Galactic rectangular coordinate frame, where the Sun is at the origin (0, 0, 0) and the Galactic Centre is at (8.178, 0, 0) kpc.

We use the GALPY package (Bovy 2015) to calculate the 3D velocity in the Galactic rectangular frame for each YSO–MC complex. The inputs are the line-of-sight velocity, the proper motions in the l and b directions (μ_l, μ_b) measured in the barycentric frame, as well as the location information, for example l, b, and d. The output is the velocity (v_x, v_y, v_z) measured in the Galactic rectangular coordinate frame.

3.2.3 *Velocities in the Galactocentric cylindrical frame and derivation of the peculiar velocity*

The Galactocentric frames have their origins at the centre of the Galaxy. To convert our velocities into these frames, we perform a rotation to align the x-axis with the mid-plane of the Galactic disc, add a translation to move the origin of our new coordinate system to the centre of the Galaxy, and subtract the solar motion relative to the Galactic Centre. Our input data include (v_x, v_y, v_z) in the Galactic rectangular frame, (x_g, y_g, z_g) in the Galactocentric rectangular frame, the solar motion $\vec{v}_\odot$ relative to the Galactic centre, $x_\odot$, and the solar height above the Galactic midplane, $z_\odot$. We adopt the solar motion ($U_\odot = 11.69\,\mathrm{km\,s^{-1}}$, $V_\odot + V_\mathrm{c} = 241.63\,\mathrm{km\,s^{-1}}$, $W_\odot = 7.67\,\mathrm{km\,s^{-1}}$) in the Galaxy (Wang et al. 2021).

The output is (v_rg, v_tg, v_zg) in the Galactocentric cylindrical frame, which represents the circular motion of the YSO–MC complex in the Galaxy. (There is a minus sign in front of v_tg, and the rotation direction is defined by the right-hand rule, corresponding to the Galaxy's clockwise rotation.)

In the Galactocentric cylindrical frame, we derive the peculiar velocity of each YSO–MC complex by subtracting the rotation motion from the tangential velocity component. The velocity rotation gradient in the solar neighbourhood is defined by $-(A+B)$ (Oort 1927), where A and B are Oort constants with values of $16.31\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ and $-11.99\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, respectively. The rotation motion is calculated as $v_{\mathrm{rotation}} = v_{\mathrm{c},\odot} - (A+B) \times (r - r_\odot)$, where r is the distance of each YSO–MC complex from the Galactic Centre, $r_\odot = 8.178\,\mathrm{kpc}$, and $v_{\mathrm{c},\odot} = 231.47\,\mathrm{km\,s^{-1}}$ (Wang et al. 2021). The peculiar velocity $\vec{v}_{\mathrm{peculiar}}$ in Galactocentric cylindrical coordinates is $(v_{\mathrm{rg}}, v_{\mathrm{tg}} - v_{\mathrm{rotation}}(r_{\mathrm{gal}}), v_{\mathrm{zg}})$ for a YSO–MC complex at a Galactic distance of r_{gal}.

3.2.4 *Velocities in the LSR and the local co-rotating frame*

The LSR frame is where the mean velocity of stars in the solar neighbourhood vanishes. The LSR frame is an inertial frame whose centre rotates with the Galaxy. It is still one of the frames where different velocity measurements are compared.

We also derive velocities in the local co-rotating frame (LCF; Li et al. 2022), which follows the mean motion of stars with an angular velocity of $v_{\mathrm{circ}}/r_{\mathrm{gal}}$, where v_{circ} is the solar circular velocity at r_{gal}, and r_{gal} is the solar distance from the centre of the Milky Way, such that the x-axis of the LCF always points to the centre of the Galaxy. This frame is non-inertial, and the Coriolis force must be accounted for. Studying the motion of stars and gas in this frame is convenient, as the angles of its axes are locked to the Galactic Centre. This frame is similar to those used in, for example, shearing box simulations of discs.

(i) **Transform velocities in the Galactic rectangular coordinate frame to the Galactocentric rectangular LSR frame** To convert velocities to the LSR frame, we account for the following. First, the Sun moves with respect to the LSR frame with a velocity of $(U_\odot, V_\odot, W_\odot)$. Second, the Sun is located 20 pc above the disc mid-plane. To correct for the solar motion, we subtract the solar velocity relative to the LSR frame: $(U_\odot = 11.69\,\mathrm{km\,s^{-1}}, V_\odot = 10.16\,\mathrm{km\,s^{-1}}, W_\odot = 7.67\,\mathrm{km\,s^{-1}})$ (Wang et al. 2021). This produces the 3D velocity of the YSO–MC complex in the Galactic rectangular LSR frame.
Next, we perform a rotation to align the xy-plane with the Galactic midplane by modifying the solar location ($x_\odot = -8.178\,\mathrm{kpc}$, $y_\odot = 0\,\mathrm{kpc}$, $z_\odot = 0.02\,\mathrm{kpc}$) (Bennett & Bovy 2019; GRAVITY Collaboration et al. 2019). The result is the 3D velocity $(v_{x,\mathrm{g}}, v_{y,\mathrm{g}}, v_{z,\mathrm{g}})$ for YSO–MC complexes in the Galactocentric rectangular LSR frame.

(ii) **Velocities in the local co-rotating frame** After deriving the 3D velocity $(v_{x,\mathrm{g}}, v_{y,\mathrm{g}}, v_{z,\mathrm{g}})$ for YSO–MC complexes in the Galactocentric rectangular LSR frame, we define a new frame, namely the local co-rotating frame (LCF) (Li et al. 2022), which rotates around the Galactic Centre with its x-axis fixed towards the centre of the Galaxy.
Compared with the LCF, the Galactocentric rectangular LSR frame rotates with an angular velocity $\vec{\Omega}_{\mathrm{frame}} = (0, 0, -\frac{v_{\mathrm{c},\odot}}{R_{\odot,\mathrm{g}}})$, where $v_{\mathrm{c},\odot} = 231.47\,\mathrm{km\,s^{-1}}$ and $R_{\odot,\mathrm{g}} = 8.178\,\mathrm{kpc}$ (GRAVITY Collaboration et al. 2019; Wang et al. 2021). The rotation speed at a point $(x_\mathrm{g}, y_\mathrm{g}, z_\mathrm{g})$ is related to the distance from the Sun, $\vec{\Delta r} = (x_\mathrm{g} + 8.178\,\mathrm{kpc}, y_\mathrm{g}, z_\mathrm{g} - 0.02\,\mathrm{kpc})$, and is calculated using $\vec{v}_{\mathrm{LSR,rotation}} = \vec{\Omega}_{\mathrm{frame}} \times \vec{\Delta r}$.
The 3D velocity in the LCF is $\vec{v}_{\mathrm{LCF}} = \vec{v}_{\mathrm{g,LSR}} - \vec{v}_{\mathrm{LSR,rotation}}$. The subtracted rotation is shown in the lower left-hand panel of Fig. 1 and is projected onto the xy-plane.

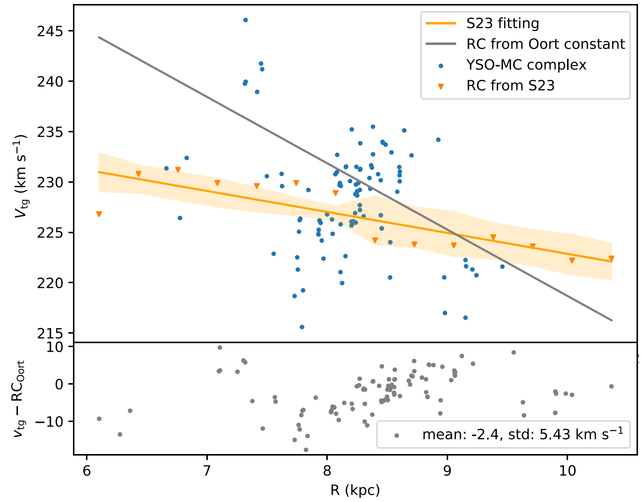

Figure 3. Top panel: Tangential velocity versus Galactic radius for YSO–MC complexes. The blue dots show the tangential velocity along the Galactic radius from our YSO–MC complexes. The orange triangles are the rotation curve data of the Milky Way derived from *Gaia* DR3 sources in Sylos Labini et al. (2023) (S23), and the orange line is the fitted rotation curve with errors. Lower panel: Differences between tangential velocities and the rotation curve derived from the Oort constant. The grey dots are the residual tangential velocity of YSO–MC complexes after the rotation curve is removed. The mean residual velocity is $-2.4\ \mathrm{km\ s^{-1}}$, and the standard deviation is $5.43\ \mathrm{km\ s^{-1}}$.

4 RESULTS

4.1 YSO–MC complex sample with 3D velocity in various reference frames

We calculate the full 3D velocities of the YSO–MC complexes in various reference frames, as illustrated in Section 3. The velocities are shown in Table 1, which includes all velocities and their corresponding errors in various reference frames. Errors are derived using GALPY, with the input errors being the proper motion of YSOs and the radial velocities from CO data.

In our sample, we highlight some well-known MCs, such as the Orion, λ Orion, Perseus, Taurus, Ophiuchus, Lupus, and Chamaeleon MCs. For those strongly associated with gas, we present their 3D velocities in Table 2.

4.2 Comparison with the Galactic rotation curve

After deriving the velocities of YSO–MC complexes in Galactocentric cylindrical coordinates, we plot their tangential velocity against their Galactic distance in Fig. 3. This figure illustrates how the rotation velocities change with the Galactocentric radius. The calculation of the tangential peculiar velocity is described in Section 3.2.3.

We compare our results with the Galactic rotation curve determined from red giant stars and *Gaia* DR3 data (Eilers et al. 2019; Wang et al. 2023), which are combined as table 1 in Sylos Labini et al. (2023) (hereafter referred to as S23). The S23 rotation curve is plotted in Fig. 3 using orange triangles and fitted with an orange line.

In the range 6.5 kpc $< R_\mathrm{g} <$ 10.5 kpc, the YSO–MC complexes show a similar decreasing trend in tangential velocity, consistent with the S23 rotation curve. This indicates that these YSO–MC complexes

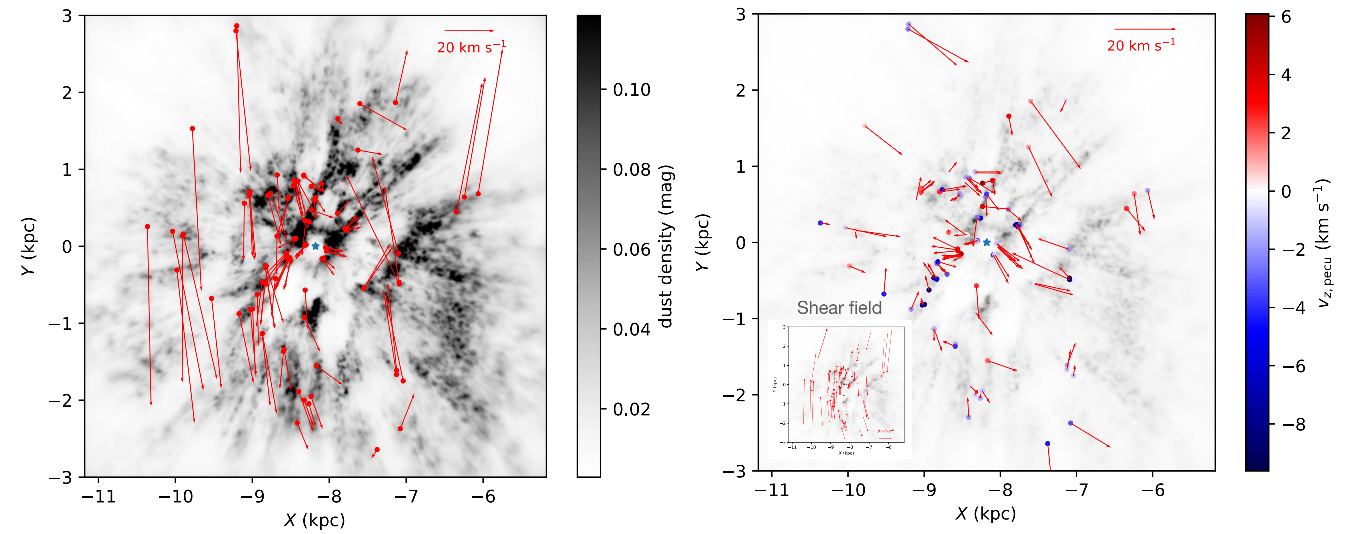

Figure 4. Left-hand panel: $\vec{v}_{\mathrm{LCF}}$ of YSO–MC complexes on the x-y plane. Red dots refer to the complex location, and the blue star at (–8.17, 0) marks the solar location. The background is the integrated dust map of the solar neighbourhood on the XY-plane from Lallement et al. (2019). Right-hand panel: 3D peculiar velocity of YSO–MC complexes on the XY-plane. The colours of the scatters represent the peculiar velocities of YSO–MC complexes along the z-axis. The inset map shows the shear field for these YSO–MC complexes. The background is the same dust map as in the left-hand panel at a different scale.

roughly follow the Galactic rotation. We then subtract the assumed rotation velocity at a certain Galactic radius, derived from the Oort constant (represented by the grey line in Fig. 3). The offsets between the rotation velocities of YSO–MC complexes and the rotation curve from Oort constants are small, with a mean value of -2.4 km s^{-1}.

In Fig. 3, we observe an increasing trend from $R_{\mathrm{g}} = 8$ kpc to $R_{\mathrm{g}} = 9$ kpc. We compare the rotation velocity of the YSO–MC complexes with the rotation velocity of 773 classical Cepheids from Mróz et al. (2019). This increasing segment aligns with part of the Cepheids data. In both observations and simulations of Milky Way structures, researchers have identified ridges in the velocity distribution, which are believed to be caused by spiral or bar structures (Antoja et al. 2018; Kawata et al. 2018; Martinez-Medina et al. 2019). The increasing part (from 8 to 9 kpc) in our sample coincides with one of the 'wiggles' located from 8 to 9 kpc in Martinez-Medina et al. (2019). Future studies combining these YSO–MC complexes with spiral arm models might help explain the features observed in the figure.

4.3 Velocity field

Our YSO–MC complex sample spans several kiloparsecs on the Galactic plane, enabling us to study the shear velocity field in the solar neighbourhood. We begin by considering their barycentric velocities in Galactic rectangular coordinates. We then subtract the frame rotation, Galactic rotation, and shear velocity to derive the peculiar velocity in Galactic rectangular coordinates.

In the top left-hand panel of Fig. 4, we plot the velocities of the YSO–MC complexes after removing frame rotation and Galactic rotation. More details about this velocity subtraction can be found in Section 3.2.4.

In the solar neighbourhood, the shear rate κ is estimated using the Oort constant A: $\kappa = 2A$ (Oort 1927). We adopt the value of the constant A as 16.31 km s^{-1} kpc^{-1} from Wang et al. (2021). When calculating the shear velocity in the Galactic plane, the solar location is used as the reference. For a given point ($\mathrm{x_g}$, $\mathrm{y_g}$) with Galactocentric distance $\mathrm{r_g} = \sqrt{(\mathrm{x_g^2} + \mathrm{y_g^2})}$, $\Delta_{\mathrm{r}} = r_g - r_{\odot,g}$. The shear velocity v_{shear} is calculated using the formula $v_{\mathrm{shear}} = \kappa \Delta_{\mathrm{r}}$. Considering the angle between the point-Sun direction and the x-axis, $\theta = \arcsin(\mathrm{y_g}/\mathrm{r_g})$, the shear velocity v_{shear} can be decomposed into two components along the x- and y-directions: $v_{x,\ \mathrm{shear}} = v_{\mathrm{shear}}\sin\theta$ and $v_{y,\ \mathrm{shear}} = v_{\mathrm{shear}}\cos\theta$. We plot the shear velocity field for each YSO–MC complex on the XY-plane in the bottom right-hand panel of Fig. 1 and in the inset panel in Fig. 4. Its absolute value ranges from 0.85 to 71.85 km s^{-1} within our sample region. The maximum shear velocity difference along the Y-axis is 130.53 km s^{-1}. The substantial shear velocity and considerable velocity difference may play an important role in shaping MCs and affect their evolution and dynamics (Elmegreen 1987; Tan 2000; Miyamoto, Nakai & Kuno 2014; Fogerty et al. 2016).

4.4 Peculiar velocity for YSO–MC complexes

We calculate the peculiar velocity for each YSO–MC complex in both Galactocentric cylindrical coordinates and Galactic rectangular coordinates. In Galactocentric cylindrical coordinates, we subtract the rotation velocity derived from the Oort constant from the tangential velocity:

$$(v_{\mathrm{rg,\ pecu}},\ v_{\mathrm{tg,\ pecu}},\ v_{\mathrm{zg,\ pecu}}) = (v_{\mathrm{rg}},\ v_{\mathrm{tg}}\ -\ v_{\mathrm{rotation}},\ v_{\mathrm{zg}}).$$

In Galactic rectangular coordinates, we subtract the frame and Galactic rotation, and the shear field:

$$(v_{x,\mathrm{pecu}},\ v_{y,\mathrm{pecu}},\ v_{z,\mathrm{pecu}}) = (v_{x,\mathrm{LCF}} - v_{x,\ \mathrm{shear}},\ v_{y,\mathrm{LCF}} - v_{y,\ \mathrm{shear}},\ v_{z,\mathrm{LCF}}).$$

The 3D peculiar velocities of the YSO–MC complexes in both coordinate systems are provided in Table 1.

In the right-hand panel of Fig. 4, we present the peculiar velocities of each YSO–MC complex on the XY-plane using arrows. The peculiar velocities along the z-axis are indicated by colours. The distributions of $v_{x,\ \mathrm{pecu}}$, $v_{y,\ \mathrm{pecu}}$, and $v_{z,\ \mathrm{pecu}}$ are plotted in the left-hand panel of Fig. 5. Along the x-axis, the complexes have a mean peculiar velocity of 2.3 km s^{-1}, with a standard deviation of 6.2 km s^{-1}. Towards the y-direction, the mean peculiar velocity is -2.4 km s^{-1}, and the standard deviation of the distribution is 5.7 km s^{-1}. The peculiar velocity along the z-axis has a mean value of -1.4

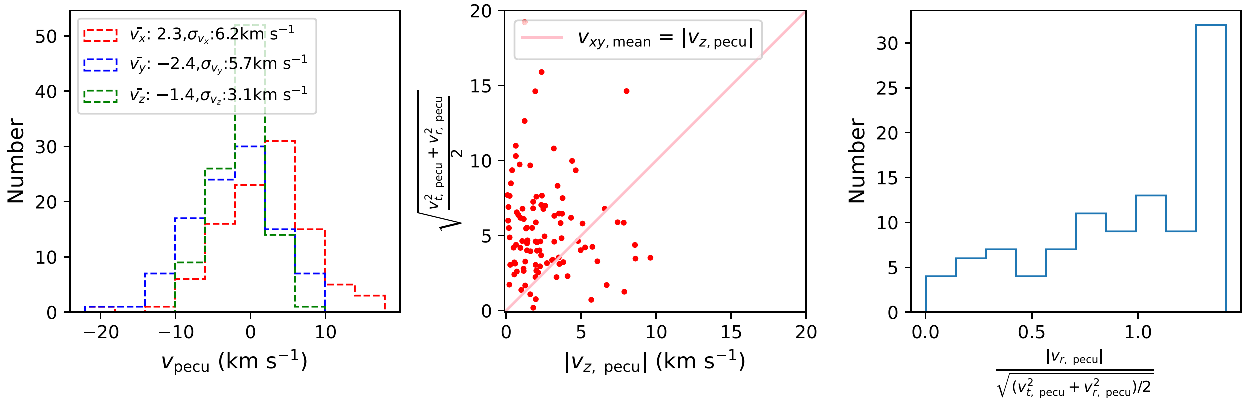

Figure 5. Left-hand panel: 3D peculiar velocity distribution of YSO–MC complexes. Red, blue, and green histograms refer to $v_{x,\,\rm pecu}$, $v_{y,\,\rm pecu}$, and $v_{z,\,\rm pecu}$, respectively. The mean velocity and standard deviation in each direction are written in the legend. Middle panel: $|v_{z,\rm pecu}|$ versus $\sqrt{\frac{v^2_{\rm r,\,pecu}+v^2_{\rm t,\,pecu}}{2}}$. The red dots show the peculiar velocity along the z-axis and the predicted peculiar velocity along the z-axis under the assumption that energy is distributed equally for the YSO–MC complexes. Right-hand panel: Distribution of the ratio between absolute radial velocity and the mean velocity on the radial and tangential planes. The mean velocity on the Galactic plane is calculated from $\sqrt{(v^2_{\rm r,\,pecu}+v^2_{\rm t,\,pecu})/2}$.

km s^{-1} and a standard deviation of 3.1 km s^{-1}. For these YSO–MC complexes, they have a mean 3D peculiar velocity of 8.6 km s^{-1}, which is consistent with the value of 5 to 10 km s^{-1} in Schulz (2005) and smaller than the typical value of about 10 km s^{-1} for maser sources in Reid et al. (2014). This value varies slightly across different works owing to different solar velocities and is heavily affected by spiral arms (Reid et al. 2009; Xu et al. 2013; Sato et al. 2014; Wu et al. 2014; Ramón-Fox & Bonnell 2018).

4.5 Anisotropic velocity

Assuming that the energy supporting the peculiar velocities is equally distributed along the radial, tangential, and z-directions, we predict the peculiar velocity $|v_{z,\,\rm pred}| = \sqrt{\frac{v^2_{\rm rg,\,pecu}+v^2_{\rm tg,\,pecu}}{2}}$, which also serves as the mean velocity on the Galactic plane. We compare the $v_{z,\,\rm pecu}$ of the YSO–MC complexes with the predicted $v_{z,\,\rm pred}$. The comparison result is plotted in the middle panel of Fig. 5. Compared with the line $y = x$, the values for $v_{z,\,\rm pecu}$ for most of the YSO–MC complexes are lower than the predicted value. This comparison result indicates an energy distribution that is lower along the z-axis than that on the Galactic plane. To study the energy distribution along the radial and tangential directions, we examine the ratio between $|v_{r,\rm pecu}|$ and the mean peculiar velocity on the Galactic plane. The ratio distribution is plotted in the right-hand panel of Fig. 5. More complexes tend to have a larger peculiar velocity along the radial direction. This observation reveals an uneven energy distribution towards the radial, tangential, and Z-directions.

5 SUMMARY

Using the 2D velocity and 3D locations of YSOs from *Gaia* DR3 and the radial velocity from the CO emission of MCs, we calculated 3D velocities for 102 YSO–MC complexes in various reference frames. All the velocities of the YSO–MC complexes and of some well-known MCs are provided in the tables. From these velocities, we derive the following results.

(i) From their tangential velocities, these YSO–MC complexes broadly follow the Galactic rotation curve. An increasing substructure at $8-9$ kpc may indicate the ridges found in Cepheid data and simulations.

(ii) Our velocity field contais significant contributions from the Galactic shear. A maximum 130 km s^{-1} velocity difference is attributed to Galactic shear.

(iii) The mean 3D peculiar velocity is 8.6 km s^{-1} for the YSO–MC complexes after subtracting the Galactic rotation and Galactic shear. This value is smaller than, yet consistent with, the peculiar velocity of 10 km s^{-1} from maser measurements.

(iv) The energy supporting the peculiar velocities is not evenly distributed along different directions. More kinetic energy is contained in the Galactic plane than along the z-axis, and more complexes tend to have higher peculiar velocities along the radial than the tangential direction .

This 3D velocity catalogue of YSO–MC complexes reveals the motion of the clouds and the shear velocity field in the solar neighbourhood. When combined with Galactic spiral models and structure simulations, it can help us understand the motion and evolution of the interstellar medium.

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from astronomical data (http://www.dendrograms.org/) and GALPY, a PYTHON package for galactic dynamics (https://docs.galpy.org/en/v1.11.0/).

DATA AVAILABILITY

This paper used published data from Marton et al. (2016) and *Gaia* DR3 (Gaia Collaboration et al. 2023). Our Tables 1 and 2 present the location and 3D velocity information in different reference frames for 102 YSO–MC complexes and some well-known MCs, respectively. Both tables will be available online upon publication.

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SUPPORTING INFORMATION

Supplementary data are available at *MNRAS* online.

3D_velocity_with_err_type1_1020.csv
readme.txt

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APPENDIX A: LIST OF ALL REFERENCE FRAMES

To clarify, we differentiate between the following frames:.

(i) **Barycentric frame:** A frame whose centre is located at the centre of gravity of the Solar system.

(ii) **Local standard of rest (LSR) frame:** The LSR frame follows the mean motion of stars in the solar neighbourhood and is centred on the Sun.

(iii) **Legacy definition of the local standard of rest (LSRK) frame:** This frame is used in radio astronomy. It contains a velocity offset from the barycentre of the solar system to remove the solar peculiar motion. It can be transformed into the barycentric frame by using an old solar velocity in Galactic coordinates: $v_{\rm LSR} = v_{\rm bary} +$

10.3cos(l)cos(b) + 15.3sin(l)cos(b) + 7.7sin(b). More information about the LSR frame can be found in Reid et al. (2009) and on two websites.[3]

(iv) **Galactic rectangular coordinate frame:** This is a rectangular coordinate system whose origin is at the barycentre of the Solar system. The positive direction of the x-axis points to the Galactic centre, and the z-axis refers to the North Galactic Pole.

(v) **Galactocentric coordinates frame:** This is a coordinate system with its origin at the Galactic Centre. It considers the precise solar location, such as the distance from the Galactic Centre and the height above the Galactic midplane, and the solar motion relative to the Galactic Centre.

(vi) **Galactocentric cylindrical coordinates:** This is a cylindrical coordinate system in the Galactocentric frame, with its origin in the Galactic Centre. The r-axis refers to the radial direction to the Galactic Centre, the t-axis refers to the tangential direction, and the z-axis refers to the North Galactic Pole.

(vii) **Galactic rectangular LSR frame:** This frame centres on the barycentre of the Solar system, with all axes aligned to the Galactic rectangular frame, and follows the mean motion of stars in the Solar neighbourhood.

(viii) **Galactocentric rectangular LSR frame:** This frame is in the local standard of rest, with all axes aligned to the Galactocentric rectangular frame. In this frame, the Sun is located at (–8.178 kpc, 0, 0.02 kpc), and the Galactic Centre is located at (0, 0, 0).

(ix) **Local co-rotating frame (LCF):** This frame has its origin at the barycentre of the Solar system. In addition to the circular motion around the Galactic Centre, it also contains a rotation around the z-axis with a velocity of the order of $v_{\rm circ}/R_{\odot}$, which allows the x-axis to be locked toward the Galactic Centre. In this frame, we can observe how the YSO–MC complex moves with respect to the local patch of material.

More descriptions about coordinates and frames can be found at https://docs.astropy.org/en/stable/coordinates/index.html#overview-of-astropy-coordinates-concepts, and more conversions can be seen at https://docs.galpy.org/en/stable/reference/util.html.

[3]https://www.atnf.csiro.au/people/Tobias.Westmeier/tools_hihelpers.php#restframes, https://science.nrao.edu/facilities/vla/docs/manuals/obsguide/modes/line

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